

FUME

Fused Unified Multi-Gas Emission Network for Livestock Rumen Acidosis Detection

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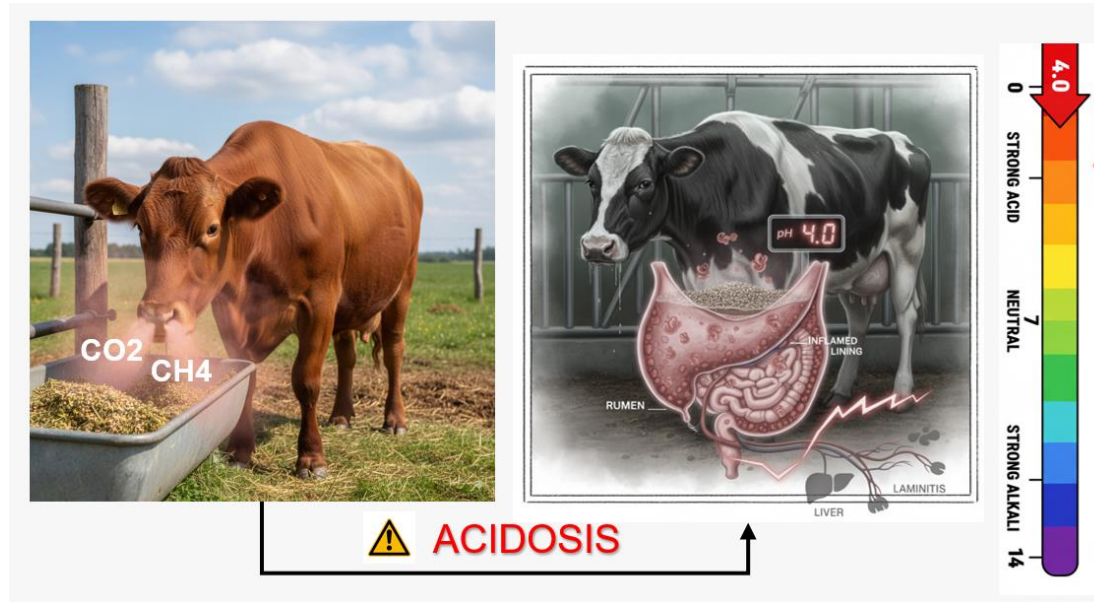
Rumen Acidosis: The Problem

19–26%

of dairy cows
affected

\$400–475

per-cow loss
per lactation



\$1B+

annual losses
in North America

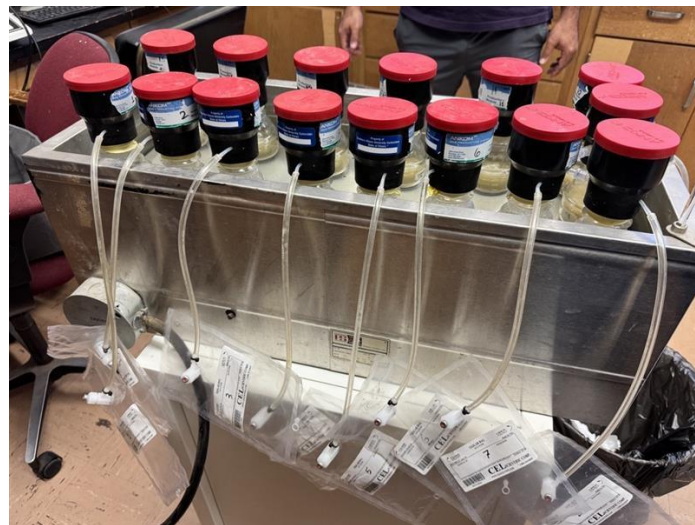
Abnormally low rumen pH (< 5.8) from rapid fermentation of high-concentrate diets

- ✗ Reduced feed intake & milk production (1–3 kg/day loss)
- ✗ Laminitis, liver abscesses, systemic inflammation
- ✗ Often undiagnosed until clinical signs appear

Dataset



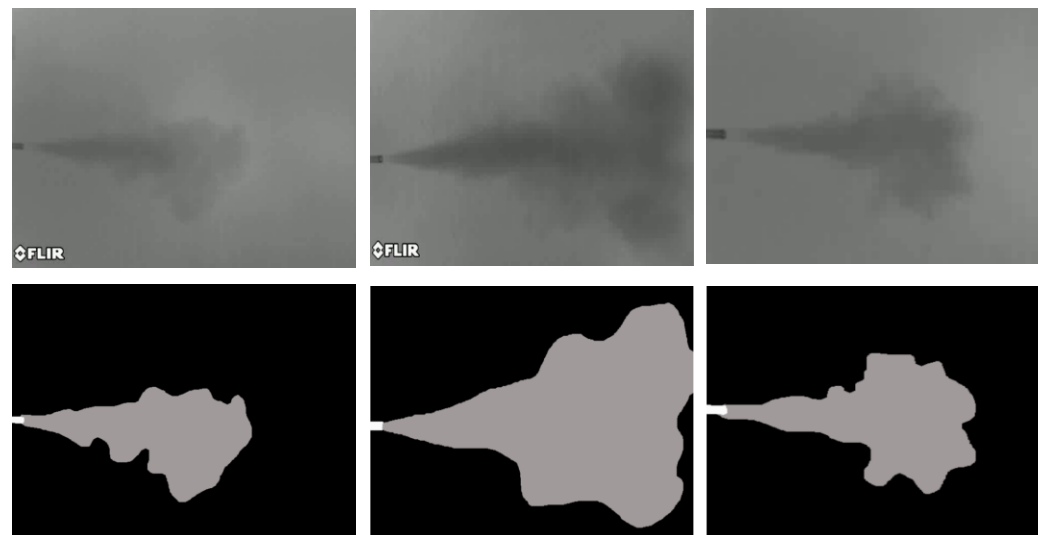
FLIR GF320 – CH4



Rumen fluid collected from fistulated cattle
↓
Fermentation substrates prepared using ANKOM RF Gas Production
↓
Samples maintained at 6 controlled pH levels (6.5, 6.2, 5.9, 5.6, 5.3, 5.0)
↓
Frames containing visible gas plumes were extracted
↓
Each frame was semi-automatically annotated with pixel-level masks



FLIR GF343 – CO2



pH	Class	Total	CO ₂	CH ₄	Train (70%)	Val (15%)	Test (15%)
6.5	Healthy	1,008	693	315	706	151	151
6.2	Healthy	1,050	721	329	735	158	157
5.9	Transitional	945	651	294	662	142	141
5.6	Acidotic	3,087	2,121	966	2,161	463	463
5.3	Acidotic	1,512	1,039	473	1,058	227	227
5.0	Acidotic	1,365	938	427	955	205	205
Total		8,967	6,163	2,804	6,277	1,346	1,344

Healthy

pH ≥ 6.0

Normal CO₂ + CH₄
balanced fermentation

Transitional

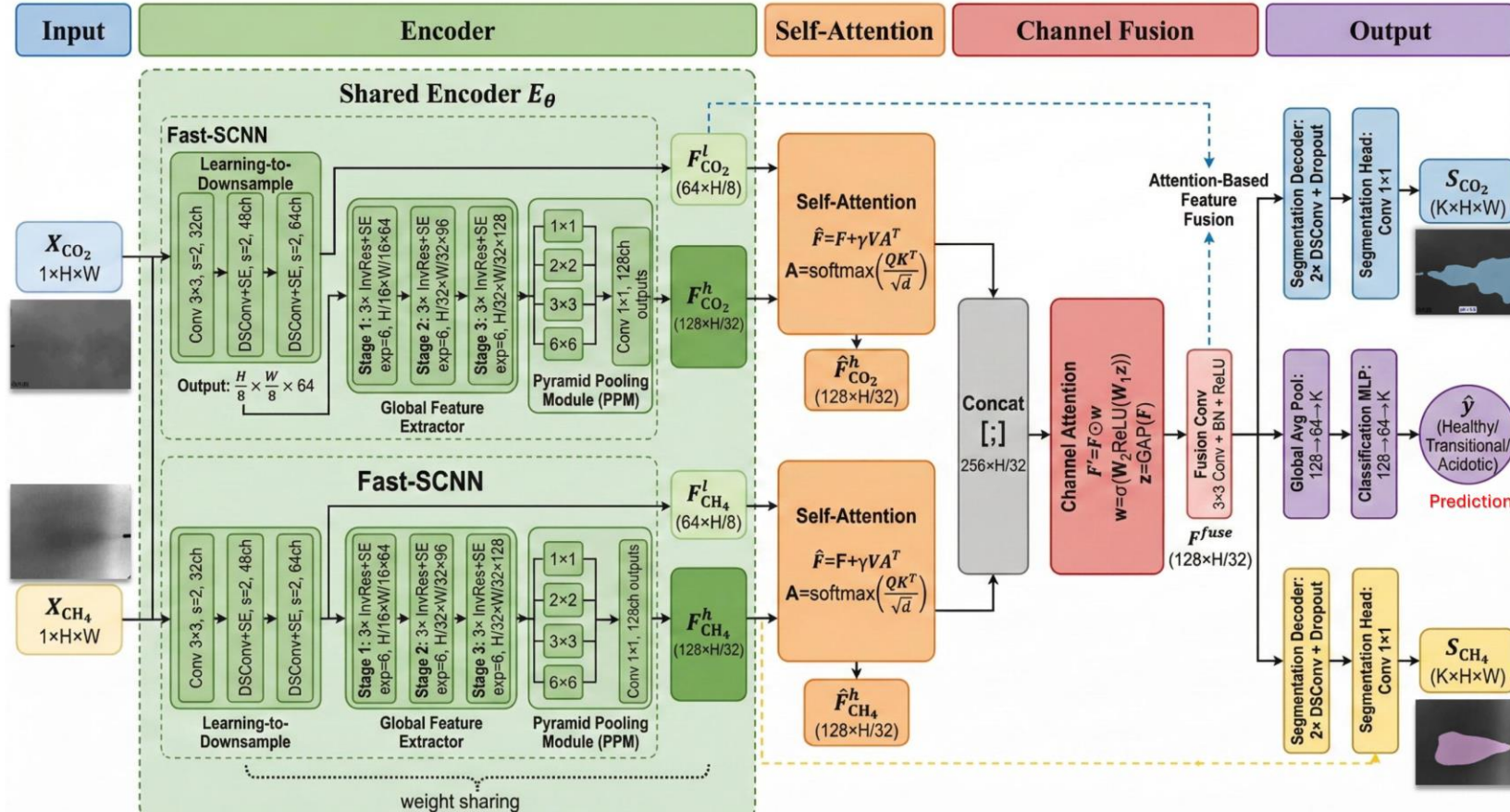
5.8 ≤ pH < 6.0

Shifting gas ratios
early warning signal

Acidotic

pH < 5.8

High CO₂, low CH₄
suppressed methanogens



Batch size: 16
Epochs: 20
Learning rate: 0.001
GPU: NVIDIA A6000

$$\text{Joint multi-task loss: } \mathcal{L} = \mathcal{L}_{seg}^{CO_2} + \mathcal{L}_{seg}^{CH_4} + \lambda \mathcal{L}_{cls}$$

$$\lambda = 0.5 \text{ for classification weight}$$

Table 1. Quantitative comparison with state-of-the-art methods. † indicates best performance. For HD95, ASD, and Latency, lower is better.

Model	Classification	Segmentation		Boundary		Efficiency			
	Acc (%)	mIoU (%)	Dice (%)	HD95 (px)	ASD (px)	Params (M)	MACs (G)	Latency (ms)	FPS
BiSeNetV2 [44]	99.04 [†]	75.91	85.85	72.13	20.89	2.77	21.01	3.54	282.49
ENet [27]	97.86	46.67	54.13	101.16	34.24	0.26 [†]	3.18	2.71	369.00
DDRNetSlim [11]	94.98	56.97	70.55	140.50	46.13	3.39	16.66	1.66 [†]	602.41 [†]
ESPNetV2 [24]	90.81	51.88	58.00	92.32	26.55	2.22	18.39	2.58	387.60
DANet [8]	48.50	47.23	54.49	103.85	34.65	1.43	5.92	2.99	334.45
RTFNet [30]	46.47	48.56	55.55	107.90	33.22	35.01	50.28	3.88	257.73
MTINet [37]	46.37	44.53	52.27	113.11	39.20	1.80	3.32	3.18	314.47
CMX [45]	33.65	26.69	29.64	187.34	61.28	4.41	13.18	1.98	505.05
Gasformer [32]	89.56	62.09	74.31	42.51	10.56	3.65	5.11	5.63	177.50
GasTwinFormer [35]	86.27	47.80	55.77	47.77	14.18	3.35	4.51	6.32	158.20
CarboFormer [14]	87.92	54.95	65.04	41.14	10.37	3.81	3.09	9.31	107.40
FUME (Ours)	98.82	80.99 [†]	89.23 [†]	46.58 [†]	14.00 [†]	1.28	1.97 [†]	3.06	326.80

Result

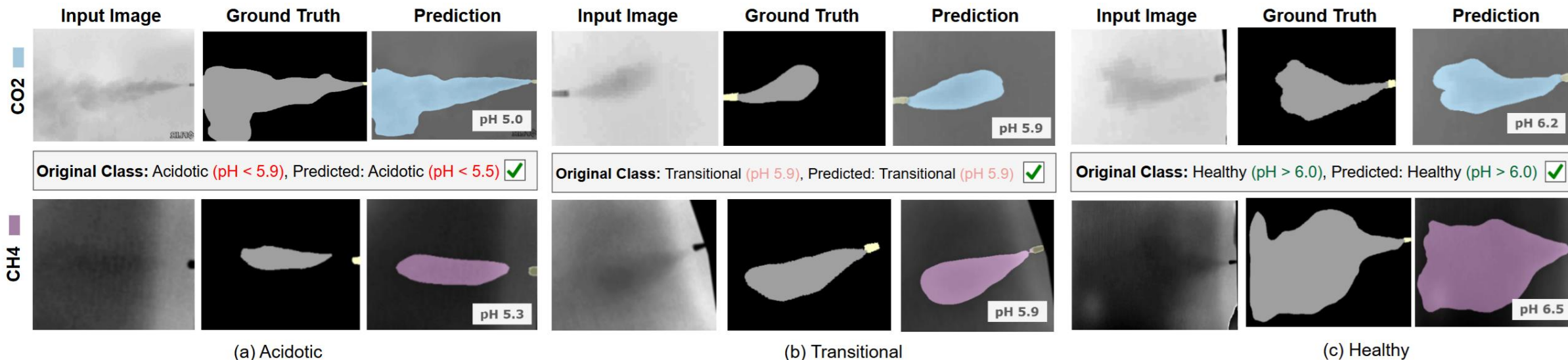


Table 4. Ablation study results. Δ denotes change relative to FUME.

Variant	Acc (%)	mIoU (%)	Dice (%)	Latency	Δ mIoU
+ Full Cross-Modal Attn	99.25	79.31	88.15	4.55	-1.68
+ Self-Attention Only	98.82	80.36	88.81	3.55	-0.63
w/o CH ₄ (CO ₂ Only)	98.61	80.52	88.92	1.78	-0.47
w/o CO ₂ (CH ₄ Only)	48.29	-	-	1.81	-
Classification Only	97.86	-	-	3.23	-
Segmentation Only	33.65	71.21	82.67	2.95	-9.78
FUME (Proposed)	98.82	80.99	89.23	3.06	-

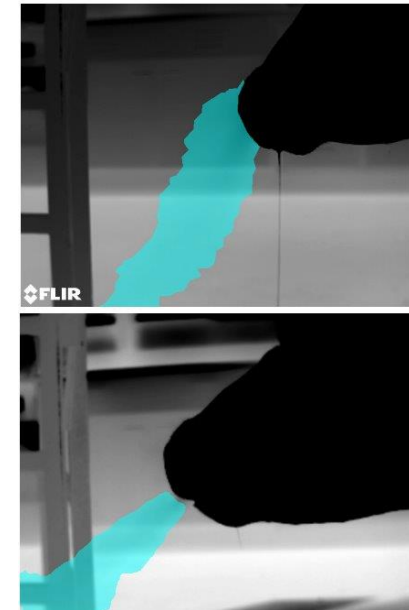
Key achievements 🏆

- ✓ Introduced **FUME**: first dual-gas emission network for rumen acidosis detection
- ✓ Created the first **dual-gas OGI dataset** ($\text{CO}_2 + \text{CH}_4$) with 8,967 annotated frames across 6 pH levels
- ✓ Achieved state-of-the-art results: **80.99% mIoU, 98.82% accuracy**
- ✓ Maintained computational efficiency: **1.28M** params, **1.97G** MACs, **326.80 FPS**
- ✓ Demonstrated CO_2 as the **primary biomarker** for acidosis; dual-task learning is essential

Future Work

- In vivo validation with live animals under farm conditions
- Temporal modeling for continuous health monitoring
- Domain adaptation from in vitro to in vivo environments
- Deploy model on edge devices for real-time farm use

Acknowledgement

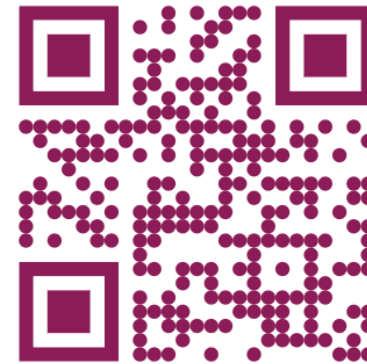


Thank You!

Any Questions?



Connect Me



FUME